Functions

Definition:

A relation $R \subseteq X \times Y$ is said to a function (mapping) from X to Y if and only if for each $x \in X$ there exists one and only one element $y \in Y$ such that $(x, y) \in R$.

Thus, a relation R is a function if and only if $(x, y) \in R$ and $(x, z) \in R$ imply $y = z$ for all $x, y, z \in X$.

i.e. R I s a function from X to Y if and only if each element $x \in X$ appears as the first coordinate in exactly one ordered pair $(x, y) \in R$.

In this case we write the function R from X to Y in the from

$$R: X \rightarrow Y$$

We will use the letters $f, g, \ldots$ instead of R.

Note that:

Any function is a relation, but the converse need not be true.

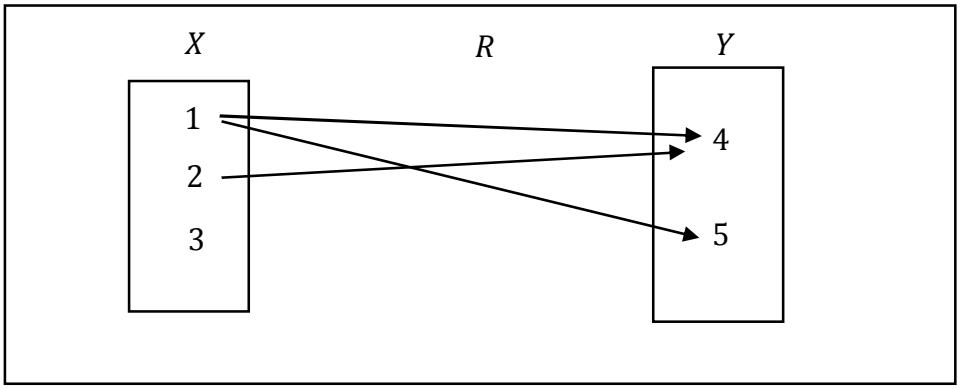
Example:

If $X = \{1, 2, 3\}$ and $Y = \{4, 5\}$, then:

1) $R = \{(1,4), (1,5), (2,4)\}$
is a relation but is not function.

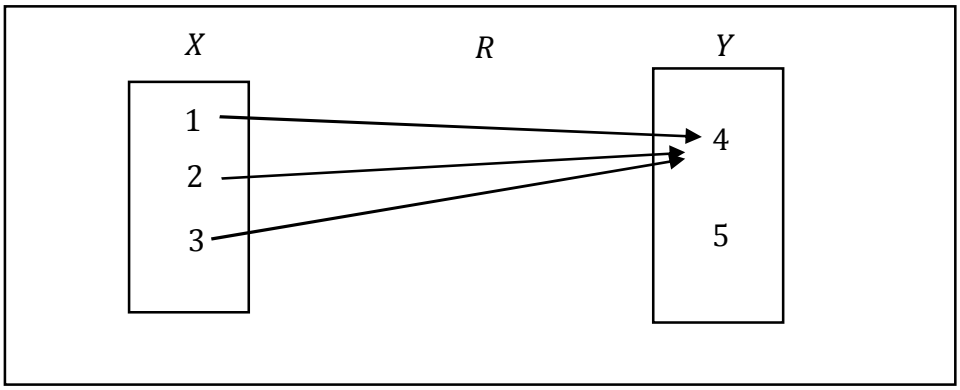
Because

i. The ordered pairs $(1,4), (1,5)$ have the same first component.
ii. $3 \in X$ but does not appear as a first component in an ordered pair of R.



2) $R = \{(1,4), (2,4), (3,4)\}$

Is a function from X to Y, although the element 4 appears as a second component in all ordered pairs of R.



Definition:

If $f: X \rightarrow Y$ is a function from X to Y, then the set X is called the domain of f, and the set Y is called the co-domain of f.

If $(x, y) \in f$, then y is called the image of x by (under) f and is denoted by $y = f(x)$.

Sometimes, y is called the value of f at x, and x is called the pre-image (inverse image) of y.

The set of all image of the elements of X is called the range of f.

i.e. Ran $f = f(X)$

Example: Let $X = \{1, 2, 3, 4\}$ and $Y = \{1, 2, 3, 4, 5\}$.

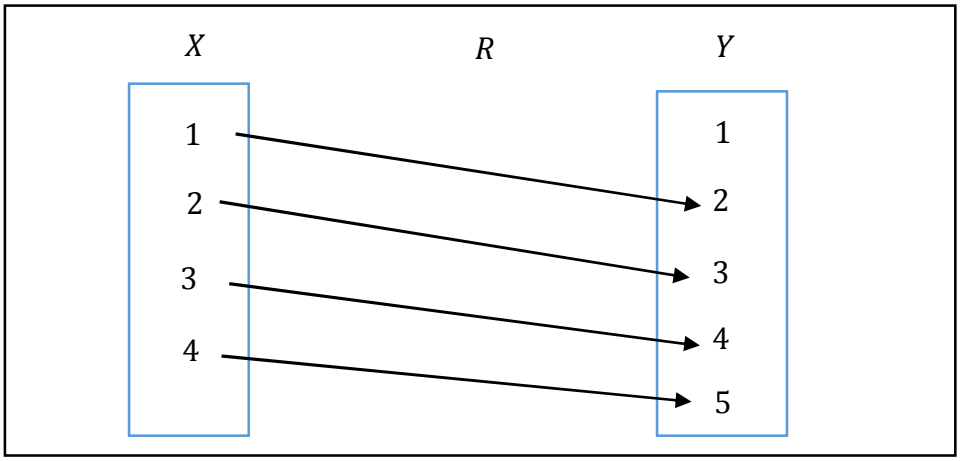
Define a relation from X to Y be

$R = \{(x, y): x \in X, y \in Y \text{ and } y = x + 1\}$

i. Express R as a set of ordered pairs.
ii. Represent R by diagrams.
iii. Show that R is a function or not, why?
iv. Find the domain, co- domain and the range of R.

Solution

$$R = \{(1,2), (2,3), (3,4), (4,5)\}$$



R is a function, because each element of X appears as a first coordinate in exactly one ordered pairs of R.

Dom $R = \{1, 2, 3,4\} = X$

Co- Dom $R = \{1, 2, 3,4,5\} = Y$

Ran $R = \{2, 3,4, 5\} \subset Y$

- Injective (one-to-one , one-one or 1-1)

Definition:

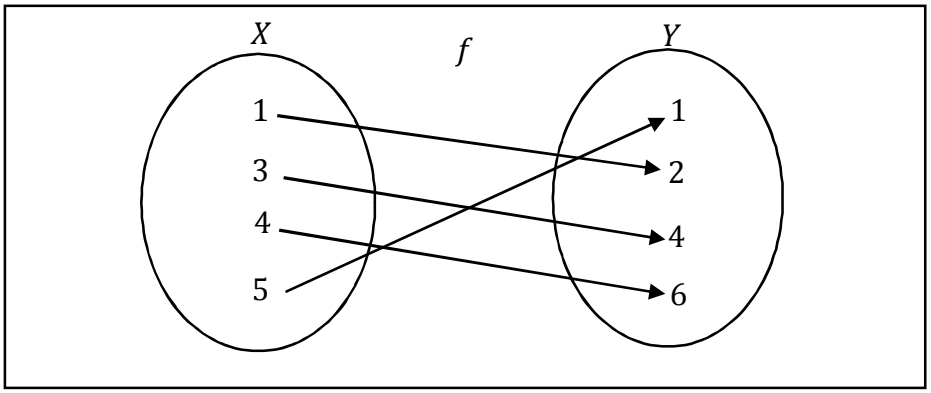
A function $f: X \rightarrow Y$ is said to be injective if it maps distinct elements in X to distinct elements in Y.

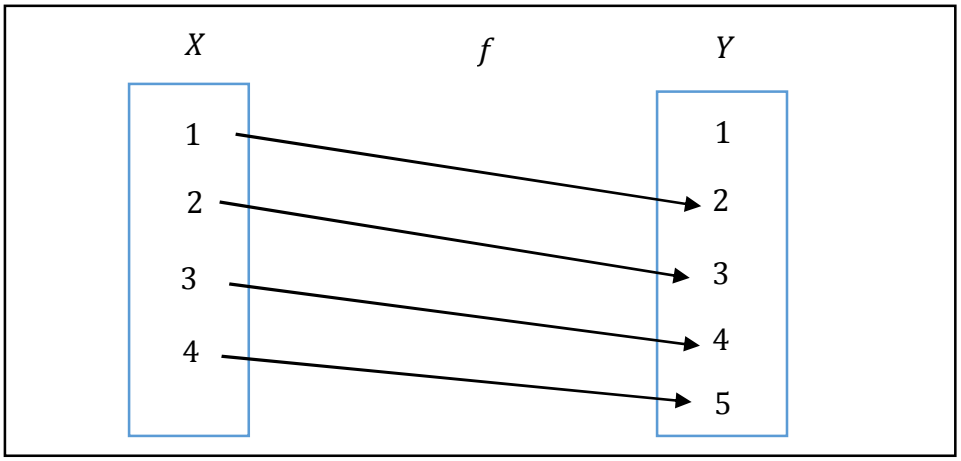
i.e. if distinct elements in X have distinct image.

i.e. $x_1 \neq x_2 \quad \Rightarrow \quad f(x_1) \neq f(x_2)$

In demonstrating the injective, it may prove to be convenient to use the following:

$f(x_1) = f(x_2) \quad \Rightarrow \quad x_1 = x_2$





Example:

Show that $f(x) = 3x - 5$ is injective.

Solution

Let $f(x_1) = f(x_2)$

$\therefore\ 3x_1 - 5 = 3x_2 - 5 \longrightarrow 3x_1 = 3x_2$

$\therefore\ x_1 = x_2$

Hence, $f(x)$ is injective.

- Subjective (onto or a surjection)

Definition:

A function $f: X \to Y$ is said to be subjective if every $y \in Y$ is the image of some $x \in X$.

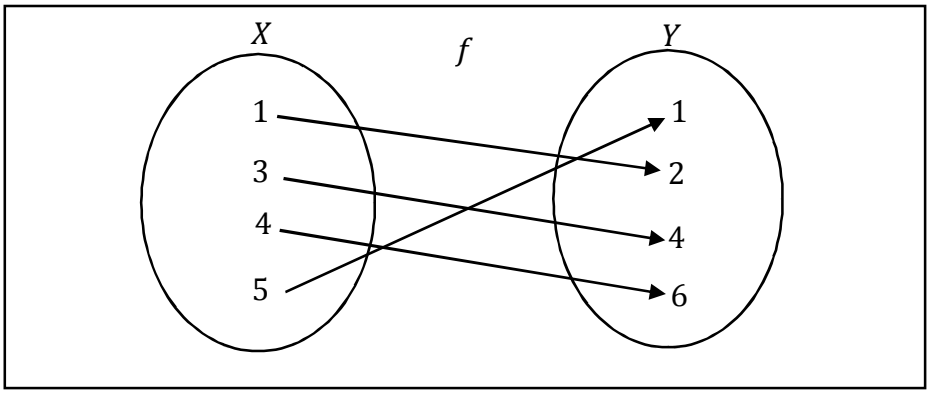
i.e.

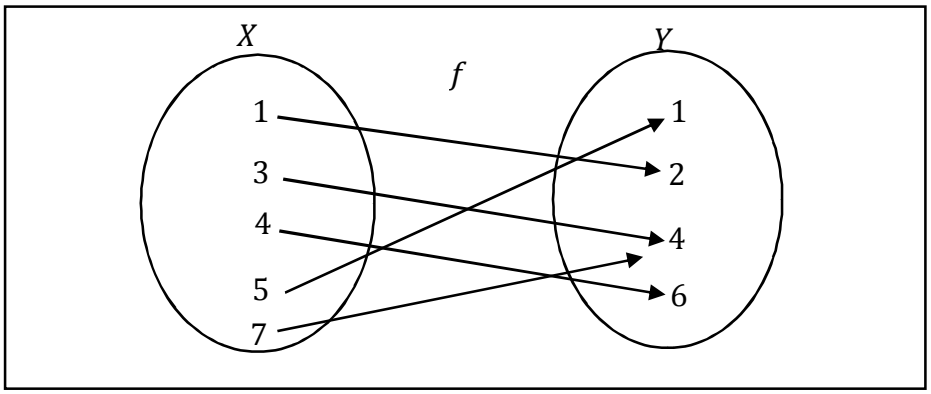
If $y \in Y \Longrightarrow \exists\ x \in X$ for which $y = f(x)$

Definition:

A function $f: X \to Y$ is surjective if and only for every y in Y, there is at least one x in X such that $f(x) = y$.

In other words f is surjective if and only if $f(X) = Y$.





- Bijective (one-to-one correspondence)

Definition:

A function $f: X \rightarrow Y$ is bijective if, for every y in Y, there is exactly one x in X such that $f(x) = y$.

Alternatively, A function $f: X \rightarrow Y$ is bijective if it is both injective and surjective.

Example: Let R be the set of real numbers

Prove that the function $f: R \to R$, defined by $f(x) = 2x + 1$, is a bijective.

Solution

i) To prove that $f(x)$ is injective,

Let $f(x_1) = f(x_2)$

$\therefore\ 2\,x_1 + 1 = 2\,x_2 + 1 \longrightarrow 2\,x_1 = 2\,x_2$

$\therefore\ x_1 = x_2$

Hence, $f(x)$ is injective.

ii) To prove that $f(x)$ is surjective,

Let $y \in R$ such that $y = f(x)$

Then $y = 2x + 1$

$\therefore\ 2x = y - 1 \longrightarrow x = \frac{1}{2}(y - 1)$

For each $y \in R$ there exists $x \in R$ such that $y = f(x)$, hence $f(x)$ is surjective.

Consequently, from (i), (ii) $f(x)$ is bijective,

Example: Let R be the set of real numbers

Determine the kind of the function

$f: R \rightarrow R$, defined by $f(x) = x^2$

Solution

i) To prove that $f(x)$ is injective,

Let $f(x_1) = f(x_2)$

$\therefore\ x_1^2 = x_1^2 \longrightarrow x_1 = \pm x_2$

$\therefore\ x_1 = x_2$

Hence, $f(x)$ is not injective.

ii) To prove that $f(x)$ is surjective,

Let $y \in R$ such that $y = f(x)$

Then $y = x^2 \longrightarrow x = \pm\sqrt{y}$

and this means that $y \geq 0$.

i.e.

Rang of $f(x) = R^+ \subset R$

Therefore $f(x)$ is not surjective.

Consequently $f(x)$ is not bijective.

- Function composition

Definition:

Let $f: X \to Y$ and $g: Y \to Z$ be two functions. Then the function $h: X \to Z$, such that $h(x) = g\big(f(x)\big)$, is called the composition function of f and g, and is denoted by:

$$h: X \xrightarrow{f} Y \xrightarrow{g} Z \qquad h = gof$$

i.e. $h(x) = (fog)(x) = f\big(g(x)\big)$

Example: Let $f, g : R \to R$ such that

$f(x) = 2x + 3$ and $g(x) = x^2$.

Find (i) $(fog)(x)$ (ii) $(gof)(x)$

Solution

(i) $(fog)(x) = f\big(g(x)\big) = f(x^2) = 2x^2 + 3$

(ii) $(gof)(x) = g\big(f(x)\big) = g(2x + 3)$

$= (2x + 3)^2 = 4x^2 + 6x + 9$

Example: Let $f, g : R \to R$ such that

$f(x) = \frac{1}{2}x - 4$ and $g(x) = 2x + 8$.

Find (i) $(fog)(x)$ (ii) $(gof)(x)$

Solution

(i) $(fog)(x) = f(g(x)) = f(x + 8)$

$= \frac{1}{2}(2x + 8) - 4 = x + 4 - 4 = x$

(ii) $(gof)(x) = g(f(x)) = g\left(\frac{1}{2}x - 4\right)$

$= 2\left(\frac{1}{2}x - 4\right) + 8 = x - 8 + 8 = x$

- Inverse function

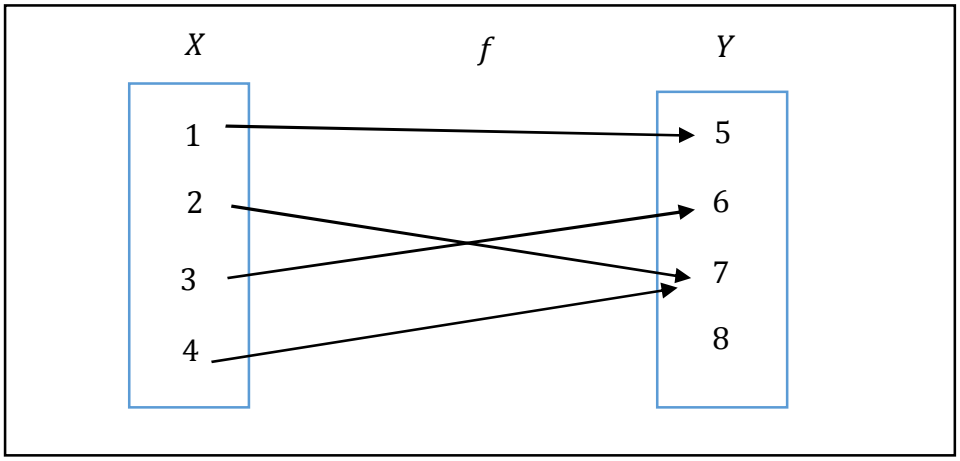
Definition:

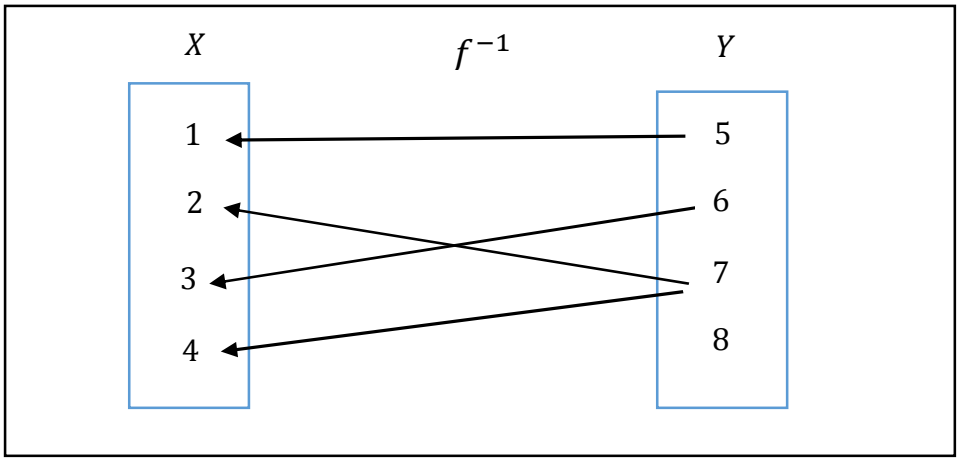
Let $f: X \to Y$ be a function of X into Y. Then $f^{-1}: Y \to X$ is called the inverse relation of f.

The inverse function of a function need not be a function as illustrated in the following example.

Example:

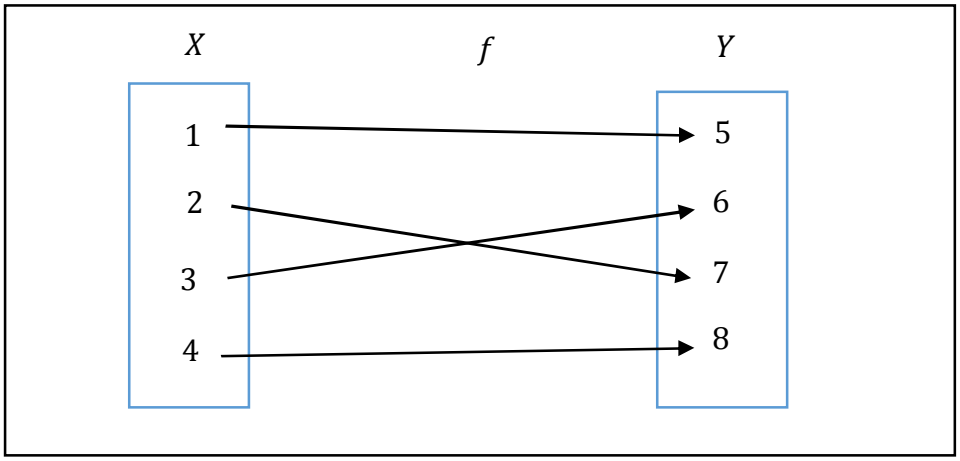
Let $f: X \to Y$ be a function defined as in the diagram.

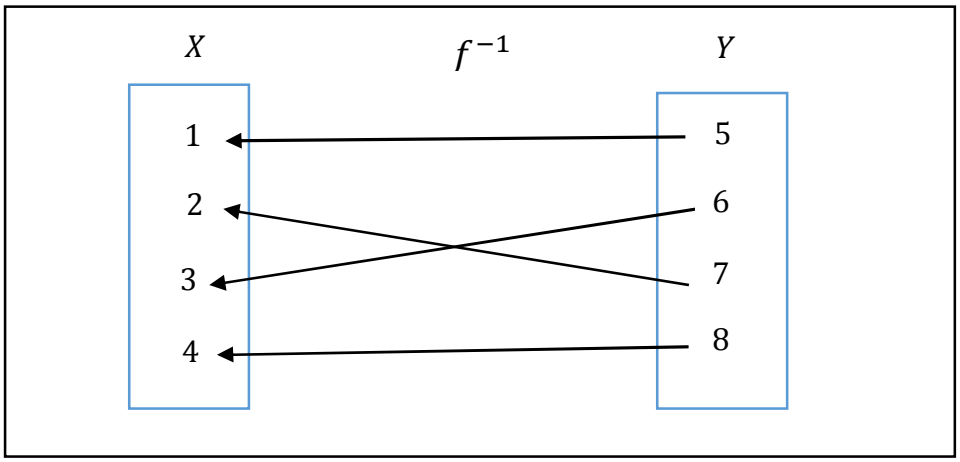




Definition:

The inverse of a function is the set of ordered pairs obtained by interchanging the first and second elements of each pair in the original function. Should the inverse of function $f(x)$ also be a function.





Steps of finding $f^{-1}(x)$

Step 1: Substitute y for $f(x)$.

Step 2: Interchange x and y.

Step 3: Solve, if possible, for y in term of x.

Step 4: Substitute $f^{-1}(x)$ for y

Step 5: Verify that $(fof^{-1})(x) = (f^{-1}of)(x) = x$.

Example:

Find the inverse of the function $f(x) = 2x - 6$.

Solution

Step 1: $y = 2x - 6$

Step 2: $y = 2x - 6 \xrightarrow{x \leftrightarrow y} x = 2y - 6$

Step 3: $2y = x + 6 \longrightarrow y = \frac{1}{2}x + 3$

Step 4: $f^{-1}(x) = \frac{1}{2}x + 3$

Step 5: $(fof^{-1})(x) = (f^{-1}of)(x) = x$.

$$(fof^{-1})(x) = f\left(f^{-1}(x)\right) = f\left(\frac{1}{2}x + 3\right)$$

$$= 2\left(\frac{1}{2}x + 3\right) - 6 = x + 6 - 6 = x$$

$$(f^{-1}of)(x) = f^{-1}\left(f(x)\right) = f^{-1}(2x - 6)$$

$$= \frac{1}{2}(2x - 6) + 3 = x$$

Exercises

1) Let $X = \{1, 2, 3, 4, 5\}$ and $Y = \{a, b, c, d, e\}$.

Determine which of the following relations is a function, then find the domain, co- domain and the range of each function.

i. $R = \{(1, e), (2, d), (3, c), (4, d), (5, a)\}$
ii. $R = \{(1, e), (2, e), (3, e), (4, b), (5, b)\}$
iii. $R = \{(2, c), (3, d), (4, d), (5, a)\}$
iv. $R = \{(1, d), (2, c), (3, e), (4, e), (5, b)\}$
v. $R = \{(a, 3), (b, 3), (c, 3), (d, 3), (e, 3)\}$
vi. $R = \{(a, 1), (c, 3), (d, 4), (e, 5)\}$

2) Which of the following relations is a function on $X = \{1, 2, 3, 4\}$. Determine the kind of each function:

i. $f_1 = \{(1,3), (2,1), (3,3), (4,3)\}$
ii. $f_2 = \{(1,4), (2,1), (3,2), (4,3)\}$
iii. $f_3 = \{(1,1), (2,3), (3,3), (4,2)\}$
iv. $f_4 = \{(1,2), (1,3), (2,4), (3,1), (4,4)\}$
v. $f_5 = \{(1,1), (2,2), (3,3), (4,4)\}$
vi. $f_6 = \{(1,4), (2,1), (4,3)\}$

3) Let $X = \{x_1, x_2, x_3\}$ and $f: X \to X$ and $g: X \to X$ be two functions such that:

$f(x_1) = x_2$, $f(x_2) = x_3$, $f(x_3) = x_1$

$g(x_1) = x_1$, $g(x_2) = x_3$, $g(x_3) = x_2$

Find each of the following;

$(fog)(x_1)$, $(fog)(x_2)$, $(fog)(x_3)$

$(gof)(x_1)$, $(gof)(x_2)$, $(gof)(x_3)$.

Is $fog = gof$?

4) If $X = R - \{5\}$, $Y = R - \{2\}$, where R is the set of all real numbers, Prove that the function $f: X \to Y$, defined by $f(x) = \frac{2x+1}{x-5}$, $x \in X$, is a bisection.

5) If $X = R - \{2\}$, $Y = R - \{3\}$, where R is the set of all real numbers, Prove that the function $f: X \to Y$, defined by $f(x) = \frac{3x+1}{x-2}$, $x \in X$, is a bijection.